

Mathemalchemy Class Activity

Course _____ Date _____ Name _____

Mathemalchemy is a travelling mathematical art installation built by 24 artists and mathematicians. Answer from what you see there; mathemalchemy.org has more.

Part 1. Codes

1. The black and white squares on the skateboarder's flag in the *Silhouettes* scene are a hidden message in binary, written in a code called **ASCII**. ASCII gives every letter, digit and punctuation mark a number, and computers store and send those numbers in **binary** (base 2), as strings of 0s and 1s. The text you type on a computer or send from your phone is still encoded this way. In ASCII, the capital letter **A** is **65**. What is 65 in binary?

2. **Cryptography** is the mathematics of protecting information. **Prime numbers** are important in cryptography because multiplying two large primes is easy, but factoring the product back into those primes is difficult, even for computers.

(a) What two numbers are the chipmunks on the playground beside the garden sorting into prime and composite?

(b) What are the squirrels doing in the garden?

3. The Cryptography Quilt has 25 panels. Choose **five**. For each, give the cipher or idea it shows, and what it is used for (or was, historically).

Cipher or idea	What it is (or was) used for

Part 2. Infinite series

4. In mathematics, what do the words **converge** and **diverge** mean?

converge: _____

diverge: _____

5. The two Ball Arches rise from the very same first ball. What do we mean when we say the short arch **converges** and the long arch **diverges**?

6. Only some of the balls on the arches are embroidered with designs. Which ones? Count the positions along the arch. Why do you think the artists chose those?

7. (a) Does the infinite sum $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$ converge or diverge?

(b) If it converges, to what value does it converge?

(c) Where does this sum appear in the exhibit?

8. An endowed scholarship fund that pays an award every year, indefinitely, is an example of a financial **perpetuity**. A dollar paid years from now is worth less than a dollar today, so each payment's **present value** (what it is worth today) is smaller than the one before. The present value of the fund is the sum of the present values of all those future payments.

(a) Is the present value of the scholarship fund a convergent or divergent series?

(b) Where in the exhibit do we find the same type of series?

Part 3. Polyhedra

9. Count the vertices, edges and faces of the five **Platonic solids**, and compute $V - E + F$ for each.

	vertices V	edges E	faces F	$V - E + F$
tetrahedron				
cube				
octahedron				
dodecahedron				
icosahedron				

10. Where in the exhibit did you find the Platonic solids?